

Unimolecular phase space theory rates by inversion of angular momentum-conserved partition function

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A simplified phase space theory (PST) of angular momentum-conserving microcanonical rate constant at specified total angular momentum J in unimolecular fragmentation under a central potential is proposed via the reverse association of fragments. Angular momentum-conserved rotational–translational sum/density of states of fragments is approximated by interpolation between “high- J ” and “low- J ” states (*Chem. Phys. Lett.*, 1996, **262**, 539), from which is obtained in closed form the corresponding J -conserved partition function $Q_{\text{xi}}(J)$; this represents the core result of this work [eqn. (20)]. A relatively simple numerical Laplace inversion routine of the product of $Q_{\text{xi}}(J)$ and the vibrational partition function accomplishes in a single stroke the inversion that leads immediately to the microcanonical rate constant $k(E, J)_{\text{PST}}$. Averaging of $Q_{\text{xi}}(J)$ over J leads directly to the canonical (thermal) PST rate constant for dissociation. The procedure is checked against available more elaborate PST results and is illustrated on cases representing five different combinations of fragment symmetries: linear + atom, sphere + atom, linear + linear, sphere + linear and sphere + sphere. The method requires minimal computational effort and is particularly efficient for calculations involving large molecules and large angular momenta.

I Introduction

A class of unimolecular reactions of considerable interest are simple bond-fissions of the type



where M is a polyatomic reactant and R_1 and R_2 are in general polyatomic fragments. In addition to the total energy, angular momentum J of reactant M is also a constant of motion that has to be conserved throughout the fragmentation process. The conservation requirement is given by the vector sum

$$J = L + J_1 + J_2 \quad (2)$$

where L is the orbital angular momentum of the two fragments about their common center of gravity, and J_1 , J_2 are the intrinsic angular momenta of the two fragments. (In the following, and unless mentioned specifically otherwise, “angular momentum” will mean the angular momentum quantum number, i.e. angular momentum in units of $\hbar = h/2\pi$, h = Planck’s constant.)

While the conservation of energy in the fragmentation of eqn. (1) is routine, conservation of angular momentum is not exactly a trivial matter. Phase space theory¹ (PST), which asserts that the rate of eqn. (1) is governed by the phase space available to products subject only to conservation of energy and angular momentum, is the earliest statistical theory that has addressed explicitly the problem of angular momentum conservation. It is now generally recognized that PST represents the statistical limit, in the sense that it produces the correct rate only if the fragmentation process is fully statistical, and in all other cases yields only an upper limit to the rate. Nevertheless PST serves as a useful benchmark which, by comparison with the experimentally observed rate, can provide information about the degree to which non-statistical effects govern a given reaction.

The development of PST described below is built upon relations for various fragment symmetries developed originally by Chesnavich and Bowers² (CB) who introduced into PST the additional constraint that the relative translational energy of the fragments under a central potential must be sufficient to clear the centrifugal barrier which serves to locate the orbiting transition state. Hence this version is sometimes referred to as the orbiting transition state (OTS) phase space theory.³

The general idea, developed in Section II, is to calculate the PST rate of eqn. (1) from the reverse association of fragments. First a J -dependent fragment sum and density of states for various symmetries is calculated by interpolation between low- and high- J states, which is then used to obtain the corresponding partition function. Inversion leads directly to the J -dependent microcanonical rate constant, and averaging over J to the canonical (thermal) rate constant. Section III describes applications to specific reaction systems representing five different fragment symmetries, Section IV discusses the results and Section V concludes.

II Theory

1 Microcanonical rate constant by detailed balance

The RRKM unimolecular rate constant for the unimolecular process of eqn. (1) (the “forward” reaction) at a given E , J can be written most generally by

$$k(E, J) = \frac{\alpha_f G^\ddagger(E, J)}{hN(E, J)} \quad (3)$$

where α_f is the (forward) reaction path degeneracy, $G^\ddagger(E, J)$ is the sum of states for the transition state at the given (E, J) and $N(E, J)$ is the corresponding density of states of reactant M .

In reactions of the type represented by eqn. (1) the vibrational potential generally has no barrier for the reverse association of fragments, so that in terms of the usual RRKM formulation the transition state for this type of process will be

“loose”, *i.e.* virtually resembling the separated fragments. Thus in the present context it becomes possible to express the forward fragmentation in terms of the reverse association of fragments (ref. 3, p. 254; ref. 4; ref. 5, p. 69).

Applying the principle of detailed balance, it can be shown⁴ that $G^\ddagger(E, J)$ is given equivalently by the convolution of the density of “internal”, *i.e.* principally vibrational, states of the separated fragments $[N_v^f(E - y)]$, and the sum of “external” rotational-translational states of the fragments $[G_x^f(y, J)]$:

$$G^*(E, J) = (2J + 1) \int_{y_m(J)}^E N_v^f(E - y) G_x^f(y, J) dy \quad (4)$$

where the energy zero is referred to the zero-point energy of the fragments, y is the sum of the rotational and relative translational energies of the fragments, and $y_m(J)$ is the J -dependent minimum y necessary to create angular momentum J . Eqn. (4) is a semi-classical expression that represents what is commonly known as the “phase space limit” since $G^*(E, J)$ is given by integrating over all the energy- and angular momentum-allowed states of separated fragments, assuming all states are equally probable. The symbols $\ddagger$ and $*$ are used here to distinguish the “traditional” and PST transition states.

$G_x^f(y, J)$, which is conceptually easier to deal with, can be represented formally by the multiple integral

$$G_x^f(y, J) = (2J + 1) \int dL \iiint dK_1 dK_2 dJ_1 dJ_2 \quad (5)$$

$y, \Delta(J, L, J_1, J_2)$

where K_1 and K_2 are the projections of J_1 and J_2 , respectively, along a fragment symmetry axis, and $y, \Delta(J, L, J_1, J_2)$ indicates that the integrals are to be evaluated for specified y and J , subject to eqn. (2). The degeneracy with respect to a space-fixed axis gives rise to the factor $(2J + 1)$ in eqn. (5) which will be dropped in the following since it appears in both the numerator and denominator of eqn. (1) and therefore cancels, unless M is a spherical top. In that case the degeneracy with respect to a space-fixed axis is $(2J + 1)^2$ so that the factor cancels only partially [*cf.* eqn. (6) below].

Eqn. (5) represents a semi-classical approximation in which angular momenta are continuous variables so that the counting of discrete states is replaced by integration. The corresponding density of states $N_x^f(y, J)$ is then the derivative of the sum $G_x^f(y, J)$ with respect to y at constant J : $N_x^f(y, J) = \partial G_x^f(y, J) / \partial y$.

If ΔH_0 is the 0 K enthalpy of the fragmentation reaction (*i.e.* the R_1 – R_2 bond energy), and if $E_r(J)$ is the rotational energy of M corresponding to J , the density of states of M in eqn. (1) is understood to be evaluated at $E + \Delta H_0 - E_r(J)$, with the energy zero as in eqn. (4); thus^{2c,4}

$$k(E, J)_{\text{PST}} = \frac{(\alpha_f/\alpha_b) G^*(E, J)}{h(2J + 1)^m N(E + \Delta H_0 - E_r(J))} \quad (6)$$

where α_b is the reaction path degeneracy for the $R_1 + R_2 \rightarrow M$ back reaction and $m = 1$ if M is a spherical top and $m = 0$ otherwise. The factor α_f/α_b amounts to the ratio of symmetry numbers of M and the fragments.⁴

The evaluation of eqn. (5) for arbitrary y and J , even with a simplified potential, is sufficiently involved² to discourage routine calculations, so that some further simplification is of practical interest.

The reduction of the problem to manageable proportions will be done in four steps: first, following CB, the interfragment vibrational potential is assumed to be a central potential of the simple form

$$V(r) = -C_n/r^n \quad (7)$$

where r is a distance along the reaction coordinate and C_n is a constant of dimensions energy $\times$ (distance) ^{n} ; second, the

density of states $N_x^f(y, J)$ is approximated by interpolation between “high- J ” and “low- J ” states; in the third step the interpolated density is used to obtain a J -conserved partition function for the fragments which then leads directly to the canonical (thermal) rate constant. Finally in the fourth step, assuming no interaction between “internal” and “external” degrees of freedom of the fragments, the product of the appropriate partition functions is inverted to obtain the sum of states $G^*(E, J)$ [eqn. (4)] directly, and from it the PST microcanonical rate constant $k(E, J)$ for specified energy E and angular momentum J [eqn. (6)].

2 The potential

The long-range attraction between two neutrals is governed mainly by the energy of dispersion for which⁶ $n = 6$ in eqn. (7). The constant C_6 may then be taken as the long-range part of the Lennard–Jones (L-J) potential, in which case $C_6 \approx 4\epsilon\sigma^6$, where ϵ is the depth of the L-J potential well and σ the collision diameter, as obtained from the viscosity or the second virial coefficient data for the reactant.

In a fragmentation reaction, however, R_1 and R_2 are radicals or atoms, *i.e.* open-shell structures, where the $1/r^6$ potential is often a poor representation of the actual interfragment potential. Therefore for the bond fissions discussed here an appropriate n and C_n were obtained from a fit to the attractive part of the Morse potential

$$V(r)_{\text{Morse}} = D_e \{1 - \exp[-\beta(r - r_e)]\}^2 \quad (8)$$

where D_e is the energy at potential minimum and the exponential parameter β is calculated from ω , the equilibrium harmonic frequency of the reaction coordinate, according to $\beta = 2\pi\omega(\mu_a/2D_e)^{1/2}$, μ_a being the reduced mass of the atoms on either side of the breaking bond. All fits were done for $r > 3$ Å, which is the region of importance in long-range interactions.

The results of the fit are given in Table 1. Not surprisingly, all powers n are non-integral, so that results that can be obtained analytically when $n = 6$, in the present instance had to be obtained numerically.

3 Reduction of $N_x^f(y, J)$

It is only in the limit of “low- J ” and “high- J ” that $G_x^f(y, J)$ [eqn. (5)] assumes simple forms which were tabulated by CB for various fragment symmetries. Table 2 gives an analogous table for the density $N_x^f(y, J)$. It was shown recently⁷ that an excellent approximation to $G_x^f(y, J)$ or $N_x^f(y, J)$ at arbitrary J can be obtained by interpolation⁸ between “low- J ” and “high- J ” forms by means of the error function:⁹

$$N_x^f(y, J) \cong N_x^f(y^\#, \text{“high } J\text{”}) \text{erf}\{\Gamma(\frac{3}{2})x\} \quad (9)$$

with x defined by

$$x = \frac{N_x^f(y^\#, \text{“low } J\text{”})}{N_x^f(y^\#, \text{“high } J\text{”})} \quad (10)$$

Here $N_x^f(y^\#, \text{“low } J\text{”}) = (2J + 1) N_x^f(y^\#, J = 0)$, and $y^\# = y - y_m(J)$. Quite analogous relations are defined⁷ for the sum of states $G_x^f(y, J)$.

The variable $y_m(J)$ is the minimum energy necessary to generate the specified angular momentum J . It can be shown²

Table 1 $-C_n/r^n$ potential from Morse fit

Reaction	n	$C_n/\text{cm}^{-1} \text{ Å}^n$
$\text{NO}_2 \leftrightarrow \text{NO} + \text{O}$	8.5	2.56×10^6
$\text{CH}_4 \leftrightarrow \text{CH}_3 + \text{H}$	6.2	8.64×10^5
$\text{NCNO} \leftrightarrow \text{NC} + \text{NO}$	5.2	5.18×10^5
$\text{CH}_2\text{CO} \leftrightarrow \text{CH}_2 + \text{CO}$	4.9	6.68×10^5
$\text{C}_2\text{H}_6 \leftrightarrow \text{CH}_3 + \text{CH}_3$	4.4	7.63×10^5

Table 2 Density of states $N(y, J)$ for various fragment symmetries

Fragment symmetry	$J = 0$	“high- J ”	r
linear + atom	$\frac{1}{2} \left(\frac{1}{yB} \right)^{1/2}$	$\frac{1}{B}$	2
sphere + atom	$1/B$	$\frac{2y^{1/2}}{B^{3/2}}$	3
linear + linear	$\frac{y^{1/2}(\sqrt{B_1} + \sqrt{B_2} - \sqrt{B_1 + B_2})}{B_1 B_2}$	$\frac{y}{B_1 B_2}$	4
sphere + linear (s) (l)	$\frac{y \sin^{-1}(B_s/(B_s + B_l))^{1/2}}{B_s^{3/2} B_l^{1/2}}$	$\frac{4y^{3/2}}{3B_s^{3/2} B_l}$	5
sphere + sphere	$\frac{4y^{3/2}}{3B_1 B_2 (B_1 + B_2)^{1/2}}$	$\frac{\pi y^2}{2(B_1 B_2)^{3/2}}$	6

B_i = rotational constant of fragment i .
 r = number of rotational degrees of freedom in the “high- J ” case.

that if $L^\dagger$ is the solution of

$$2B_r(J - L) - \frac{2n}{n-2} \frac{L^{(n+2)/(n-2)}}{b_n^{2n/(n-2)}} = 0 \quad (11)$$

where $B_r = B_1 B_2 / (B_1 + B_2)$ is the effective rotational constant of the two fragments having individual rotational constants B_1 and B_2 , $y_m(J)$ is given by

$$y_m(J) = \left(\frac{L^\dagger}{b_n} \right)^{2n/(n-2)} + B_r (L^\dagger - J)^2 \quad (12)$$

With μ = reduced mass of fragments, we have

$$b_n = \frac{1}{h} \left(\frac{2n\mu}{n-2} \right)^{1/2} \left[\frac{(n-2)C_n}{2} \right]^{1/n} \quad (13)$$

which is a constant that connects relative translational energy (E_t) to maximum orbital angular momentum: $L = b_n E_t^{(n-2)/2n}$.

4 The J -conserved partition function

Using the interpolated density $N_x^f(y, J)$ of eqn. (9) we could in principle calculate the corresponding “exact” J -conserved partition function $Q_x(J)$ for “external” states (k = Boltzmann’s constant) as

$$Q_x(J) = \int_{y_m(J)}^{\infty} N_x^f(y, J) e^{-y/kT} dy \quad (14)$$

which represents the usual definition of a partition function except that it recognizes explicitly that there is a minimum energy $y_m(J)$ required to generate the specified angular momentum. While the actual numerical evaluation of $Q_x(J)$ in eqn. (14) poses no problem in principle, it is still computationally intensive even for J -interpolated $N_x^f(y, J)$.

Only in the case of linear fragment + atom is it possible to obtain $Q_x(J)$ directly from eqn. (14), using the J -interpolated density $N_x^f(y, J)$. From Table 2 and eqns. (9) and (10) we have (B = rotational constant of the linear fragment)

$$N_x^f(y, J) = \frac{1}{B} \operatorname{erf} \left\{ \left(J + \frac{1}{2} \right) \frac{1}{2} \left(\frac{\pi B}{y} \right)^{1/2} \right\} \quad (15)$$

With this expression for the density eqn. (14) can be evaluated analytically:¹⁰

$$Q_x(J) = e^{-y_m(J)/kT} \frac{kT}{B} \left[1 - \exp \left\{ - \left(J + \frac{1}{2} \right) \left(\frac{\pi B}{kT} \right)^{1/2} \right\} \right] \quad (16)$$

In all other, less tractable cases, a compact, and at the same time reasonably accurate, approximation to the ‘exact’ partition function $Q_x(J)$ at arbitrary J can be obtained by a similar interpolation, but this time between “low- J ” and “high- J ” forms of the actual partition function $Q_x(J)$. With the approx-

imation $Q_x(\text{“low-}J\text{”}) \cong (2J + 1)Q_x(J = 0)$ the two forms are easily obtained analytically by integration from eqn. (9) using for $N_x^f(y, J)$ the data in Table 2. For $J = 0$ we have

$$Q_x(J = 0) = c_0 \left(\frac{r-3}{2} \right)! (kT)^{(r-1)/2} \quad (17)$$

where c_0 stands for the constants in front of the various powers of y in the $J = 0$ column of Table 2, and r , which gives the number of rotational degrees of freedom in the “high- J ” case, is given in the last column of the Table. Observe that in each case $r - 1$ is the number of so-called transitional degrees of freedom in the reactant M, *i.e.* degrees of freedom that become rotations in the fragments. If c_h represents the constants in front of the various powers of y in the “high- J ” column of Table 2, the partition function for the “high- J ” “external” states is

$$Q_x(\text{“high-}J\text{”}) = c_h \left(\frac{r-2}{2} \right)! (kT)^{r/2} \quad (18)$$

The J -conserved partition function $Q_{xi}(J)$ for “external” states at arbitrary J can then be approximated by interpolation (hence the additional subscript “i”):

$$Q_{xi}(J) = e^{-y_m(J)/kT} Q_x(\text{“high-}J\text{”}) \times \operatorname{erf} \left\{ \Gamma \left(\frac{3}{2} \right) \frac{(2J + 1)Q_x(J = 0)}{Q_x(\text{“high-}J\text{”})} \right\} \quad (19)$$

Using r and the explicit forms of the constants c_0 and c_h for the various fragment symmetries from Table 2, eqn. (19) can be put in the more compact form

$$Q_{xi}(J) = e^{-y_m(J)/kT} b (kT)^{r/2} \operatorname{erf} \left\{ \left(J + \frac{1}{2} \right) \frac{a}{(kT)^{1/2}} \right\} \quad (20)$$

where the parameters a , b and $r/2$ are given in Table 3 for each fragment combination.

One way to appreciate the accuracy of the above approximation to $Q_{xi}(J)$ is to compare $Q_{xi}(J)$ calculated from eqn. (20) with the “exact” partition function calculated the “hard way” from eqns. (5) and (14) by executing all the numerical integrations [see eqn. (22) in ref. 2(a)]. This is done in Fig. 1 which shows excellent agreement.

5 The canonical rate constant

With eqn. (20) it is possible to write directly the J -resolved canonical, *i.e.* thermal high-pressure PST rate constant $k_\infty(J, T)_{\text{PST}}$ for dissociation at specified angular momentum J and

Table 3 Parameters in the J -interpolated partition function of eqn. (20)

Fragment symmetry	a	b	$r/2$
linear + atom	$\frac{\pi}{2} B^{1/2}$	$\frac{1}{B}$	1
sphere + atom	$B^{1/2}$	$\frac{\sqrt{\pi}}{B^{3/2}}$	$\frac{3}{2}$
linear + linear	$\frac{\pi}{2} (\sqrt{B_1} + \sqrt{B_2} - \sqrt{B_1 + B_2})$	$\frac{1}{B_1 B_2}$	2
sphere + linear (s) (l)	$B_1^{1/2} \sin^{-1} \left(\frac{B_s}{B_s + B_l} \right)^{1/2}$	$\frac{\sqrt{\pi}}{B_s^{3/2} B_l}$	$\frac{5}{2}$
sphere + sphere	$\left(\frac{B_1 B_2}{B_1 + B_2} \right)^{1/2}$	$\frac{\pi}{(B_1 B_2)^{3/2}}$	3

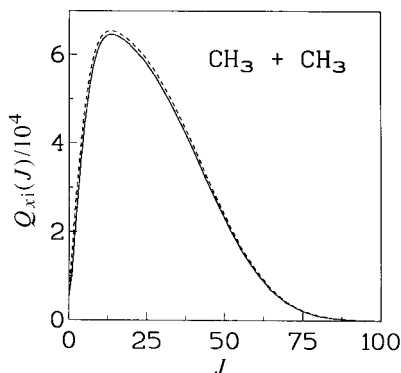


Fig. 1 Comparison of the “exact” 300 K partition function calculated from eqn. (14) using “exact”^{2b} $N_x^f(y, J)$ (solid line), with that calculated from the interpolated formula in eqn. (20) (dashed line). Both curves refer to the $\text{CH}_3 + \text{CH}_3$ system, each radical spherically averaged, *i.e.* using the geometric average of methyl radical rotational constants; thus $B_1 = B_2 = (9.45^2 \times 4.73)^{1/3} = 7.503 \text{ cm}^{-1}$. $Q_{xi}(J)$ in the vertical legend stands for $Q_{xi}(J)$ or $Q_x(J)_{\text{exact}}$.

specified temperature T :

$$k_{\infty}(J, T)_{\text{PST}} = \frac{kT}{h} \frac{\alpha_f}{\alpha_b} g_e \frac{Q_{vf} Q_{xi}(J)}{Q_M} e^{-\Delta H_0/kT} \quad (21)$$

where Q_{vf} is the partition function for the internal degrees of freedom of the fragments and $Q_M = Q_v Q_r$ is the total partition function for the reactant molecule factorized into vibrational (Q_v) and rotational (Q_r) parts. The factor g_e is the ratio of effective electronic degeneracies of the fragments and the reactant. Averaging of $Q_{xi}(J)$ over all J produces the experimentally observable T -dependent thermal rate constant:

$$k_{\infty}(T)_{\text{PST}} = \frac{kT}{h} \frac{\alpha_f}{\alpha_b} g_e \frac{Q_{vf} \langle Q_{xi}(J) \rangle_J}{Q_M} e^{-\Delta H_0/kT} \quad (22)$$

Here $\langle Q_{xi}(J) \rangle_J$ is the average of $Q_{xi}(J)$ over the thermal distribution of angular momenta $P(J) = (2J + 1)^m \exp[-J(J + 1)B_e/kT]$, with B_e the doubly or triply degenerate equilibrium rotational constant of a symmetric ($m = 1$) or spherical ($m = 2$) top reactant, respectively. The normalization constant of $P(J)$ is implicitly included in Q_M as the rotational partition function Q_r . The corresponding high-pressure recombination rate constant then follows by application of the equilibrium constant.

6 The microcanonical rate constant

Another way to check the approximation represented by eqn. (20) is to invert $Q_{xi}(J)$ to obtain the J -conserved sum or density of “external” states $G_x^f(E, J)$ or $N_x^f(E, J)$ and compare it with the “exact” result [*e.g.* eqns. (11)–(13) in ref. 7]. This comparison is made in Fig. 2, again with satisfactory results, which thus completes the demonstration of the validity of this route to the corresponding density or sum of “external” states.

However the principal advantage of this inversion is that at virtually no additional computational effort it is easy to calculate the sum of internal – “external” states $G^*(E, J)$ directly by inverting the product $Q_{vf} Q_{xi}(J)$. This obviates the calculation of the convolution integral in eqn. (4) and leads directly to the PST microcanonical rate constant $k(E, J)$ [eqn. (6)]. The inversion is readily accomplished numerically by the method of steepest descents⁵ (S-D), described briefly below, which produces a smooth-function approximation to $G^*(E, J)$.

The general S-D procedure is to define for the sum or density of states at total energy E a function $\phi(\beta)$ ($\beta = 1/kT$) which is the logarithm of the integrand in the inversion integral.⁵ Writing $Q_{vf}(\beta)$ for Q_{vf} , and $Q_{xi}(J, \beta)$ for $Q_{xi}(J)$ to show the β -dependence explicitly, the integrand in the present

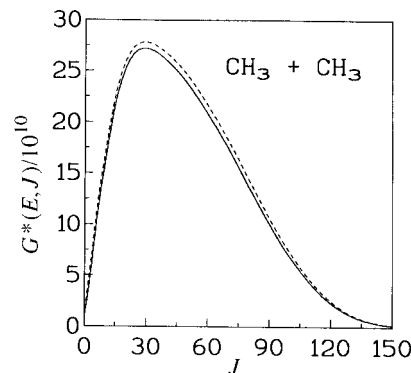


Fig. 2 Comparison of the sum of states for internal – “external” states $G^*(E)/(2J + 1)$ [eqn. (4)] at 10^4 cm^{-1} as a function of J . The solid line is “exact”, *i.e.* obtained by actual numerical integration of eqn. (4) using “exact” $G_x^f(y, J)$. The dashed line is obtained by S-D inversion of the interpolated partition function of eqn. (20). $G^*(E, J)$ in the vertical axis legend is an abbreviation for $G^*(E, J)/(2J + 1)$. The system is the spherically averaged $\text{CH}_3 + \text{CH}_3$ (see caption of Fig. 1 for parameters).

instance, assuming independence of internal and “external” degrees of freedom, is the product of three factors $Q_{vf}(\beta)Q_{xi}(J, \beta) \exp(\beta E)$. Thus

$$\phi(\beta) = \ln Q_{vf}(\beta) + (E - y_m(J))\beta + \ln b - \left(\frac{r}{2} + k\right) \ln \beta + \ln \text{erf}(\alpha\beta^{1/2}) \quad (23)$$

where $k = 0$ selects the density, $k = 1$ the sum of states, and $\alpha = (J + \frac{1}{2})\alpha$; the other symbols are those in eqn. (20). If we let

$$\frac{\partial \ln \text{erf}(\alpha\beta^{1/2})}{\partial \beta} = F(\beta) = \frac{\alpha e^{-\alpha^2 \beta}}{(\pi\beta)^{1/2} \text{erf}(\alpha\beta^{1/2})} \quad (24)$$

the first derivative is

$$\frac{\partial \phi(\beta)}{\partial \beta} = \frac{\partial \ln Q_{vf}(\beta)}{\partial \beta} + (E - y_m(J)) - \frac{k + \frac{r}{2}}{\beta} + F(\beta) \quad (25)$$

and the second is

$$\phi''(\beta) = \frac{\partial^2 \phi(\beta)}{\partial \beta^2} = \frac{\partial^2 \ln Q_{vf}(\beta)}{\partial \beta^2} + \frac{k + \frac{r}{2}}{\beta^2} - F(\beta) \left(\alpha^2 + \frac{1}{2\beta} + F(\beta) \right) \quad (26)$$

If β^* is the solution of $\partial \phi(\beta)/\partial \beta = 0$, the result for the sum or density of states, to first-order, is the standard expression which represents a smooth-function approximation:

$$\left. \begin{matrix} N(E, J) \\ k=0 \\ G(E, J) \\ k=1 \end{matrix} \right\} = \frac{e^{\phi(\beta^*)}}{(2\pi\phi''(\beta^*))^{1/2}} \quad (27)$$

The logarithmic derivatives of the quantum form of the vibrational partition function $Q_{vf}(\beta)$ are readily calculated in the separable harmonic oscillator or Morse oscillator approximation, including hindered internal rotors, if necessary.¹¹

The density of states of the reactant in the denominator of eqn. (6) follows by a S-D routine similar to the above by simply substituting $Q_v(\beta)$ for $Q_{vf}(\beta)$ and deleting terms related to $Q_{xi}(J, \beta)$. On the other hand, if the reactant involves also an active K-rotor (see below the discussion of NCNO), the S-D

routine simply proceeds as above with all terms related to $Q_{\text{vi}}(J, \beta)$ included, but with $y_{\text{m}}(J) = 0$ and parameters a , b and r redefined as discussed in the Appendix.

It was verified that averaging over E and J of $k(E, J)_{\text{PST}}$ obtained by inversion gives back in each case the correct $k_{\infty}(T)_{\text{PST}}$.

III Applications

The PST procedure outlined above for microcanonical and canonical calculations will be illustrated on fragmentations representative of the five principal classes of fragment symmetries as listed in Tables 2 and 3.

The calculated PST microcanonical rate constant for dissociation [eqn. (6)], using the inversion procedure based on eqn. (20), will be compared with experimental and/or calculated angular momentum-resolved microcanonical rate constants available in the literature. The calculations used for comparison are either by one of the incarnations of the Variational Transition State Theory (VTST), or (at low value of J only) by "exact" PST, meaning in essence the quantum version of eqn. (5) evaluated by actual summations.

The calculated PST thermal rate constants for dissociation and recombination, the latter *via* eqn. (22) and the equilibrium constant, are listed in Table 4 for the five reaction systems. These may be compared with available experimental data, from which one can obtain in principle the factor $\Phi_r(T) = k_{\infty}(T)_{\text{exp}}/k_{\infty}(T)_{\text{PST}}$ as a measure of the deviation of the PST result from the "true" rate constant. Since $k_{\infty}(T)_{\text{exp}}$ generally requires an extrapolation to high pressure, experimental data are subject to uncertainty, particularly as regards the temperature coefficient. In this respect experimental recombination results are probably preferable since they are subject to smaller error owing to their smaller temperature coefficient. Note that the factor $\Phi_r(T)$ is the same whether defined in terms of a ratio of decomposition or recombination rate constants.

All potential parameters, shown in Table 1, were obtained by a Morse fit as described in Section II-2. It turned out that the results are only weakly sensitive to the values of n and C_n . Equilibrium constants were calculated from molecular parameters of the species involved and checked against data in JANAF tables¹² where available. The three-dimensional fragment rotors CH_2 and CH_3 were approximated as spheres. Discussion of reactions representing the various symmetry classes follows.

1 Linear fragment + atom

An example is $\text{NO}_2(1^2\text{A}_1) \leftrightarrow \text{NO}(2^2\Pi_{\text{g}}) + \text{O}(3^4\text{P}_{\text{J}})$. Due to spin-orbit interactions, $\Omega = \frac{1}{2}, \frac{3}{2}$ and $J = 0, 1, 2$, so that there are 18 adiabatic potential energy curves, each doubly degenerate, connecting reactant and products, of which only the lowest with $\Omega = \frac{1}{2}$ and $J = 2$ leads to the ground state reactant;¹³ thus $g_{\text{e}} = 5$. The potential parameters for NO_2 in Table 1 represent an approximate fit to the lowest potential energy curve

shown in Fig. 7 of ref. 13. Basically this is a Morse fit shifted to larger distances by 0.17 Å. Data for the electronic levels of the reactant and products were taken from JANAF tables;¹² $\alpha_{\text{f}}/\alpha_{\text{b}} = 2$. Anharmonic density of states of reactant was calculated in the Morse approximation using diagonal anharmonicity coefficients.¹⁴ It has been shown¹⁵ that this is actually a very good approximation to the exact density of anharmonic coupled oscillators.

Fig. 3 shows the energy dependence of $G^*(E, J)/(2J + 1)$ [eqn. (4)] for this reaction at several J (solid lines) calculated by inversion from the J -conserved partition function *via* eqn. (16), using data from Tables 1 and 3. Molecular and fragment parameters were taken from Table A1 of ref. 16. Included for comparison are results (symbols) calculated by the Flexible Transition State Theory (FTST), taken from Fig. 5 in ref. 16 [their $N_{\text{EJ}}/(2J + 1)$ is $G^*(E, J)/(2J + 1)$ in the present notation]. Fig. 4 shows the result of the present PST calculation of the energy dependence of the actual rate constant $k(E, J)$ at four different values of J obtained by inversion of the J -conserved partition function [eqn. (20)].

There are two recent extensive reviews of experimental recombination data, which both agree that $k_{\infty}(T)_{\text{exp}} = 3 \times 10^{-11} \text{ cm}^3 \text{ molec}^{-1} \text{ s}^{-1}$ at 300 K, but disagree on the temperature coefficient: one¹⁷ gives $k_{\infty}(T)_{\text{exp}} = 3 \times 10^{-11} (T/298)^{-0.75}$ (200–2500 K), in the same units, while the other¹⁸ gives the rate as temperature-independent at 3×10^{-11} throughout the range 200–1500 K. Consequently with the help of Table 4 only $\Phi_r(300 \text{ K}) = 0.82$ can be reported with some uncertainty.

2 Spherical fragment + atom

The example here is $\text{CH}_4 \leftrightarrow \text{CH}_3 + \text{H}$ if the rotational constants of the methyl radical are spherically averaged. In this

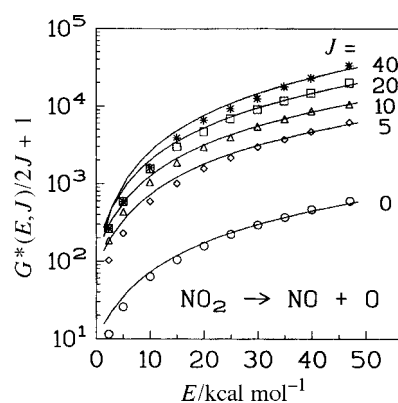


Fig. 3 Semi-logarithmic plot of the J -conserved sum of states $G^*(E, J)/(2J + 1)$ for the decomposition $\text{NO}_2 \rightarrow \text{NO} + \text{O}$ at five different values of J and at energies between 0 and 50 kcal mol⁻¹. Symbols are FTST calculations (ref. 16, Fig. 5), solid lines are results of the present PST calculation by inversion of the J -conserved partition function [eqn. (16)].

Table 4 Thermal PST rate constants calculated from eqn. (22). Recombination: $k_{\infty}(T) = C(T/298)^n$, 300–500 K. Dissociation: $k_{\infty}(T) = A_{\infty} \exp(-E_{\infty}/kT)$, 1500–2000 K

Reaction	$C/\text{cm}^3 \text{ molec}^{-1} \text{ s}^{-1}$	n	A_{∞}/s^{-1}	$E_{\infty}/\text{cm}^{-1}$
$\text{NO}_2 \leftrightarrow \text{NO} + \text{O}$	3.65×10^{-11}	0.09	3.00×10^{15}	25282
$\text{CH}_4 \leftrightarrow \text{CH}_3 + \text{H}$	2.52×10^{-9}	0.88	2.12×10^{18}	37715
$\text{NCNO} \leftrightarrow \text{NC} + \text{NO}$	4.11×10^{-11}	0.28	2.97×10^{15}	15925
$\text{CH}_3\text{CO} \leftrightarrow \text{CH}_3 + \text{CO}$	1.25×10^{-10}	0.47	1.44×10^{16}	29378
$\text{C}_2\text{H}_6 \leftrightarrow \text{CH}_3 + \text{CH}_3$	2.70×10^{-10}	0.49	7.0×10^{17}	30439

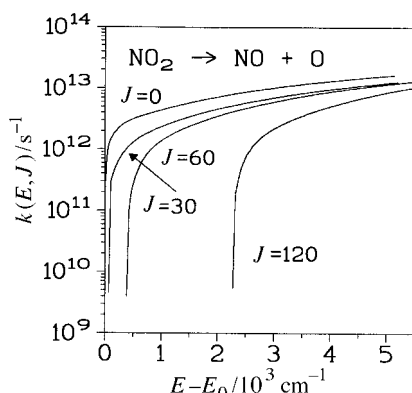


Fig. 4 Semi-logarithmic plot of the rate constant $k(E, J)$ for the decomposition $\text{NO}_2 \rightarrow \text{NO} + \text{O}$ over the range 0–5000 cm^{-1} and at $J = 0, 30, 60$ and 120 , calculated by inversion of the J -conserved partition function [eqn. (16)]. E_0 stands for ΔH_0 .

case the standard Morse $\beta = 1.72$ appears to give a poor representation of the long-range part of the $\text{CH}_3\text{--H}$ potential. The values shown in Table 1 were obtained from a fit to a Morse potential with $\beta = 2.2$ recommended¹⁹ as the best representation of an *ab initio* potential. Molecular parameters for CH_4 and CH_3 were taken from ref. 20. Methane reactant was treated as a spherical top, $\alpha_t/\alpha_b = 2$. Fig. 5 shows the present PST results for the rate constant $k(E, J)$ obtained by inversion of the J -conserved partition function [eqn. (20)] at $J = 0$ and 50 (solid lines). For comparison are shown variational results²¹ (symbols): asterisks for $J = 0$ and triangles for $J = 50$. On the scale of the graph results for intermediate J would be poorly resolved.

A recent review of thermal data²² (based on ref. 23) gives for recombination $k_\infty(T)_{\text{exp}} = 3.5 \times 10^{-10} \text{ cm}^3 \text{ molec}^{-1} \text{ s}^{-1}$, temperature-independent in the range 300–1000 K. Two older reviews give for $k_\infty(T)_{\text{exp}}$ a negative temperature coefficient: $2.05 \times 10^{-10} (T/298)^{-0.4}$ (300–2500 K) [ref. 24] and $4.2 \times 10^{-10} (T/298)^{-0.85} \exp(-173/T)$ (300–600 K) [ref. 25] in the same units. These results, with the help of Table 4, yield $\Phi_r(300 \text{ K}) \approx 0.08\text{--}0.14$.

3 Linear fragment + linear fragment

A much-studied example is $\text{NCNO} \leftrightarrow \text{NC} + \text{NO}$. Experimental²⁶ and calculated data by FTST²⁷ for the microcanonical rate constant $k(E, J)$ are available only for $J = 3$ at

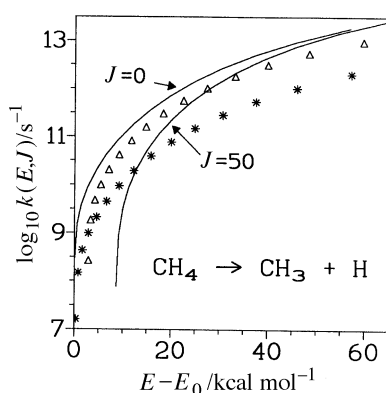


Fig. 5 Semi-logarithmic plot of the microcanonical rate constant $k(E, J)$ for the decomposition $\text{CH}_4 \rightarrow \text{CH}_3 + \text{H}$ as a function of $E - E_0$ ($=\Delta H_0$). The labeled solid lines are the present PST results for $J = 0$ and 50 . Symbols are a variational²¹ calculation: asterisks, $J = 0$; triangles, $J = 50$. On the scale of the graph any PST or variational data between $J = 0$ and $J = 50$ would be poorly resolved.

energies below 800 cm^{-1} . NCNO parameters for the present calculation were taken from refs. 27 and 28; potential parameters are in Table 1. In this case $\alpha_t/\alpha_b = 1$.

NCNO is essentially a prolate symmetric top (two smallest rotational constants almost equal). It was assumed in the present calculation that the rotational quantum number K is not conserved, *i.e.* that the rotational energy of the K-rotor contributes to density of states $N(E, J)$ of the reactant. In order to prevent a serious overcount, this contribution to $N(E, J)$ must correctly reflect the $-J \leq K \leq +J$ constraint which is particularly severe at low values of J , as in the present case. It is shown in the Appendix that this restriction leads to a K-rotor partition function that, except for different parameters, has the form of eqn. (20), and therefore requires little modification of the S-D inversion routine for microcanonical calculations as set out in eqns. (23)–(27) with $k = 0$.

Fig. 6 compares the present calculation of the microcanonical rate constant for the singlet decomposition at $J = 3, 30$ and 60 (solid lines) with data at $J = 3$ from Fig. 5 of ref. 27 identified as “quantum PST” (circles), which for the present purpose may be considered “exact” PST. $J = 30$ is the average J for this system at 300 K. Fig. 6 includes also experimental rate constants²⁶ at $J = 3$ (crosses).

Experimental thermal recombination data are available from two sources: a review²⁹ gives $k_\infty(T)_{\text{exp}} = 6.6 \times 10^{-11} \text{ cm}^3 \text{ molec}^{-1} \text{ s}^{-1}$, temperature-independent in the range 500–1500 K, and a more recent experiment³⁰ gives $k_\infty(T)_{\text{exp}} = 3.7 \times 10^{-11} (T/298)^{-0.3} \text{ cm}^3 \text{ molec}^{-1} \text{ s}^{-1}$ in the range 99–450 K. From Table 4 we have $\Phi_r(300 \text{ K}) \approx 0.45\text{--}0.8$, the higher value obtained on the assumption that the review data are also valid at 300 K.

4 Spherical fragment + linear fragment

A specific example is $\text{CH}_2\text{CO} \leftrightarrow {}^1\text{CH}_2 + \text{CO}$, also dealt with by FTST.³¹ Insofar as the present calculation is concerned, this case is similar to the decomposition of NCNO in that the reactant molecule is practically a symmetric top and experimental data are available only³² for $J = 3$. On the assumption that the rotational quantum number K is not conserved, a treatment of the K-rotor similar to the preceding was adopted in the calculation of the reactant density of states.

The CH_2 fragment is an asymmetric rotor with three unequal rotational constants which were spherically averaged for the purpose of this calculation; $\sigma_t/\sigma_b = 1$. Molecular data for reactant and products were taken from ref. 31, using³³ $\Delta H_0 = 30116 \text{ cm}^{-1}$. Potential data are in Table 1.

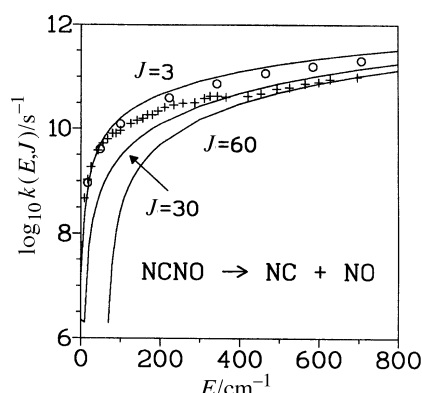


Fig. 6 Semi-logarithmic plot of the rate constant $k(E, J)$ for the decomposition $\text{NCNO} \rightarrow \text{NC} + \text{NO}$ between 0 and 800 cm^{-1} . The crosses (+) are experimental data,²⁶ circles (O) is the “quantum PST” calculation [ref. 27, Fig. 5], all at $J = 3$. The solid lines represent results of the present PST calculation by inversion of the J -conserved partition function at $J = 3, 30$ and 60 .

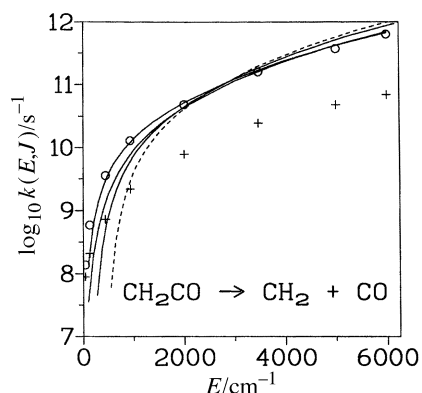


Fig. 7 Semi-logarithmic plot of the rate constant $k(E, J)$ for the decomposition $\text{CH}_2\text{CO} \rightarrow \text{CH}_2 + \text{CO}$ between 0 and 6000 cm^{-1} . Crosses (+) are experimental results³⁶ and the circles (O) represent a “quantum PST” calculation [ref. 31, Fig. 2], all at $J = 3$. The solid lines are the present PST result obtained by inversion of the J -conserved partition function. From left to right, the curves are for $J = 3, 21$ and 75 ; the dashed line is for $J = 100$.

Fig. 7 compares the present calculation of the microcanonical rate constant at $J = 3$ (leftmost solid line) with the PST calculation (circles) shown in Fig. 2 of ref. 31, considered as “exact” for the present purpose. In this case it may be seen that the inversion of the J -resolved partition function at $J = 3$ yields a rate constant that is practically identical to the “exact” result. The Figure also includes present PST results for higher values of J , in particular $J = 21$ which is the average J for this system at 300 K. Clearly the microcanonical rate constant has a rather weak J -dependence. As in the case of NCNO, experimental results (crosses) do not agree with PST at $J = 3$.

A review²⁴ gives for recombination $k_{\infty}(T)_{\text{exp}} < 10^{-15} \text{ cm}^3 \text{ molec}^{-1} \text{ s}^{-1}$ in the 300–2500 K range as the upper limit. This is almost five orders of magnitude smaller than the PST value (Table 4) and suggests that the experiment and PST calculation may not refer to the same reaction.

5 Two spherical fragments

The example here is $\text{C}_2\text{H}_6 \leftrightarrow \text{CH}_3 + \text{CH}_3$ if the rotational constants of the methyl radicals are spherically averaged as

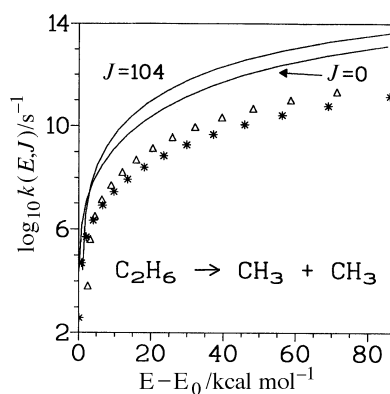


Fig. 8 Semi-logarithmic plot of the microcanonical rate constant $k(E, J)$ for the decomposition $\text{C}_2\text{H}_6 \rightarrow \text{CH}_3 + \text{CH}_3$ as a function of $E - E_0 (= \Delta H_0)$ (See caption of Fig. 1 for parameters). The curves labeled $J = 0$ and $J = 104$ are the present PST results calculated by inversion of the J -conserved partition function [eqn. 20]. Symbols are variational results²¹ for $J = 0$ (asterisks, *) and $J = 104$ (triangles, Δ). The latter are identical to FTST results [ref. 16, Fig. 8] for $J = 100$. On the scale of the graph any PST or variational data between $J = 0$ and $J = 100$ would be poorly resolved.

in the decomposition of methane. Molecular data were taken from ref. 34 and potential parameters are in Table 1; $\alpha_t/\alpha_b = \frac{1}{2}$. It was assumed that the rotational quantum number K of ethane is not conserved, so that a treatment of the K-rotor similar to the cases of NCNO and CH_2CO was adopted in the calculation of C_2H_6 density of states.

The labeled curves in Fig. 8 are the present PST results at $J = 0$ and $J = 104$, calculated by inversion of the J -conserved partition function [eqns. (20)–(27)]. For comparison are shown variational results²¹ for $J = 0$ (asterisks) and $J = 104$ (triangles). The latter are identical to the FTST $J = 100$ result (Fig. 8 in ref. 16).

Recent thermal recombination data all agree on a negative temperature coefficient: one review³⁵ gives $k_{\infty}(T)_{\text{exp}} = 5.4 \times 10^{-8} T^{-1.10} \exp(-160/T) \text{ cm}^3 \text{ molec}^{-1} \text{ s}^{-1}$ and another³⁶ gives $k_{\infty}(T)_{\text{exp}} = 8.78 \times 10^{-11} \exp(-T/723)$ in the same units (200–1000 K). At 300 K both yield practically the same value for $k_{\infty}(T)_{\text{exp}}$ (5.97 and 5.80×10^{-11} , respectively), so that from Table 4 we have $\Phi_r(300 \text{ K}) = 0.21$ – 0.22 . These data were used in preference to older results.^{37,38}

IV Discussion

The fragmentations of NCNO and CH_2CO involve molecules that are small enough that at low J the present microcanonical PST routine for $k(E, J)$ offers little advantage with respect to the “exact” PST; thus the comparison of “present” and “exact” PST at $J = 3$ serves principally to document the validity of the present approach. This is shown in Fig. 6 and especially in Fig. 7 which both show good agreement between the “exact” and “present” version of PST at $J = 3$. In both fragmentations PST’s produce rate constants somewhat higher than experiment, which indicates that the reaction is some distance from the statistical limit as represented by PST. In the fragmentation of NCNO this would militate in favor of $\Phi_r(300 \text{ K}) \approx 0.45$ (rather than 0.8 which seems clearly unreasonable). For future reference Figs. 6 and 7 also include calculated microcanonical rate constants up to $J = 60$ for NCNO and up to $J = 100$ for CH_2CO .

In the case of the fragmentation of NO_2 there are both low- and high- J data available for comparison, as shown in Fig. 3. The close agreement between the PST and FTST sum of states at $J = 0, 5, 10, 20$ and 40 indicates that the reaction should not be far from the statistical limit, which is supported by the thermal result $\Phi_r(300 \text{ K}) \approx 0.8$. The actual rate constant $k(E, J)$ plotted in Fig. 4 can be usefully compared with $k(E, J)$ in Fig. 2 of ref. 39 calculated from the Statistical Adiabatic Channel (SACM) model.

The fragmentations of C_2H_6 and CH_4 illustrate the present method for other fragment symmetries, more complex molecular species and high angular momenta, for which it was in fact specifically designed. The efficiency of the routine for the calculation of $k(E, J)$ is unaffected by the number of vibrational degrees of freedom, the value of the energy E or angular momentum J , and therefore is particularly suitable for such cases.

Fig. 5 compares a variational rate constant $k(E, J)$ for the fragmentation of CH_4 with the present PST calculation at $J = 0$ and $J = 50$. PST yields for $J = 50$ a rate constant with a particularly large offset to higher energies that makes the PST result at low energies look lower than the variational one. This is because the threshold energy $y_m(J)$ tends to be large, on account of the very small reduced mass of the fragments [$\mu = 0.938 \text{ amu}$]. The PST microcanonical rate constant for the decomposition of C_2H_6 (Fig. 8), on the other hand, does not show quite as large an offset with respect to a variational calculation since here the fragment reduced mass has a more normal value [$\mu = 7.5 \text{ amu}$]. It seems therefore that eqns. (12) and (13) may not be accurate enough for very low μ .

Note also in Table 4 the large calculated pre-exponential factor A_∞ for the dissociation of CH_4 , indicative that the calculated PST “transition state” is very loose indeed. However experiment suggests that the reaction must have a tighter transition state, far from the PST limit, which is consistent with the very low value of Φ_r (300 K).

While experimental data generally produce a zero or negative temperature coefficient for recombination, Table 4 shows that all PST thermal recombination constants have a positive temperature coefficient. This is because in PST there is no tightening of the “transition state” at high energies as in a variational routine. In fact on examination the PST thermal recombination rate constant $k_\infty(T)_{\text{PST}}$ has a rather simple structure, which can be approximated to within better than 10% in the range 300–500 K, taking $Q_{\text{vf}} \approx 1$:

$$k_\infty(T)_{\text{PST}} \approx \text{const.} \frac{\langle \exp[-y_{\text{m}}(J)/kT] \rangle_J}{(kT)^{1/2} \mu^{3/2}} \quad (28)$$

where μ is the reduced mass of the fragments and const. includes, apart from numerical factors, reaction path and electronic degeneracies. The average $\langle \exp[-y_{\text{m}}(J)/kT] \rangle_J$ is understood to be over the distribution $P(J)$ specified above in connection with eqn. (22). This approximation identifies $k_\infty(T)_{\text{PST}}$ for recombination as depending principally on μ , and through $P(J)$ and $y_{\text{m}}(J)$ on the potential and the rotational constants of the reactant and products.

For a symmetric top reactant it turns out that $\langle \exp[-y_{\text{m}}(J)/kT] \rangle_J \approx kT$, while for a spherical top (methane), $\langle \exp[-y_{\text{m}}(J)/kT] \rangle_J \approx kT^{3/2}$, so that approximately $k_\infty(T)_{\text{PST}}$ is proportional to $T^{1/2}$ for the former, and proportional to T for methane; this is borne out by the data in Table 4.

For the recombination $\text{O} + \text{NO}$ the temperature dependence of $k_\infty(T)_{\text{PST}}$ for recombination is more complicated because there intrudes the temperature dependence of electronic partition functions for NO and O which depend on temperature roughly as $1/T$; thus $k_\infty(T)_{\text{PST}}$ is likely to have a small temperature coefficient, as confirmed in Table 4.

Since experimental recombination constants have a zero or negative temperature coefficient, it follows that $\Phi_r(T)$ always decreases with temperature, *i.e.* that the PST overestimate of rate not only gets larger as temperature increases but also increases approximately with increasing number of degrees of freedom of the reactant, CH_4 excepted (*cf.* exponent n in Table 4). On the microcanonical level, this implies that the overestimate of $k(E, J)$ increases with energy E and the size of the reactant. Owing to the uncertainty regarding the exact temperature coefficient of experimental recombination constants, it is unprofitable to speculate about the exact temperature dependence of $\Phi_r(T)$.

The factor $\Phi_r(T)$ used here is similar to the “rigidity factor” $f_{\text{rigid}}(T)$ of Troe⁴⁰ who has interpreted it as a measure of the anisotropy of the potential. The difference between $\Phi_r(T)$ and $f_{\text{rigid}}(T)$ is merely that the latter involves a PST limit calculated from SACM by taking $\alpha/\beta \rightarrow \infty$, where α is an adjustable SACM “looseness” parameter and β is the standard Morse β . For $\text{CH}_3 + \text{H}$ the reported value is ^{40b} $f_{\text{rigid}}(300 \text{ K}) \approx 0.20$, which is not far from the present $\Phi_r(300 \text{ K}) = 0.14$, both using the same temperature-independent $k_\infty(T)_{\text{exp}} = 3.5 \times 10^{-10} \text{ cm}^3 \text{ molec}^{-1} \text{ s}^{-1}$. In another publication,⁴¹ after recalculation using the same experimental data as in Section III, we find $f_{\text{rigid}}(300 \text{ K}) \approx 0.76$ for $\text{NO} + \text{O}$, in fair agreement with $\Phi_r(300 \text{ K}) = 0.82$ found here, and for $\text{CH}_3 + \text{CH}_3$ the result is $f_{\text{rigid}}(300 \text{ K}) \approx 0.2$, which happens to be also the present result.

V Conclusion

A general procedure is set up for the calculation of simplified phase space theory (PST) rate constants that conserve angular

momentum, assuming central potential and separability of vibrational and rotational–translational degrees of freedom.

The core and novel result is the J -conserved partition function $Q_{\text{xi}}(J)$ [eqn. (20)] which is obtained in closed form expression for five principal classes of fragment symmetries, arbitrary total angular momentum J and arbitrary total energy y of the “external” degrees of freedom. This is accomplished by approximating $Q_{\text{xi}}(J)$ using interpolation between “high- J ” and “low- J ” forms.

The principal interest of $Q_{\text{xi}}(J)$ is that a relatively simple numerical Laplace inversion of the product of $Q_{\text{xi}}(J)$ with the fragment vibrational partition Q_{vf} accomplishes in a single stroke the determination of the sum of states $G^*(E, J)$ which then leads directly to the microcanonical rate constant $k(E, J)_{\text{PST}}$ [eqn. (6)]. This not only ensures that angular momentum is conserved insofar as the “external” degrees of freedom are concerned, but also makes redundant the actual calculation of the convolution integral [eqn. (4)]. In a second pass by the same routine, by merely redefining some of the constants, and substituting the reactant vibrational partition function Q_{v} for Q_{vf} , one obtains the density of states of the reactant $N(E)$ [the denominator in eqn. (6)], with or without active symmetric top K-rotor (see Appendix). Since the same inversion routine serves for both $G^*(E, J)$ and $N(E)$, the whole procedure is quite efficient, and this for any size reactant, regardless of the value of E or J .

The canonical (thermal) rate constant $k_\infty(T)_{\text{PST}}$ is obtained by averaging of $k(E, J)_{\text{PST}}$ over E and J , or, considerably more directly, by averaging $Q_{\text{xi}}(J)$ over J . Obviously $k_\infty(T)_{\text{PST}}$ generates no new information (in fact it contains less) but connects PST with the more abundant source of experimental thermal data and serves as a consistency check on $k(E, J)_{\text{PST}}$. Thus $Q_{\text{xi}}(J)$ is the central element of the present approach that provides access to both the microcanonical and canonical rate constants.

The routines are first tested against other more elaborate microcanonical PST calculations on the fragmentations of NCNO and CH_2CO at low angular momentum and low energies, as well as against a variational calculation on the fragmentation of NO_2 at larger energies and angular momenta, with encouraging results that show a general usefulness of the method. The method is then applied to the decomposition of two large molecules, CH_4 and C_2H_6 , at high angular momenta, as an example of the cases for which the efficiency of the present method would be most useful, and for which it was primarily designed. The chosen examples cover five principal classes of fragment symmetries.

The canonical PST rate constants for recombination calculated here all have a positive temperature coefficient and a fairly simple structure which shows them to depend principally on the reduced mass of the fragments, on the potential and on the rotational constants of the reactant and fragments. These recombination rate constants may be compared with available experimental data, from which one could obtain in principle the factor $\Phi_r(T) = k_\infty(T)_{\text{exp}}/k_\infty(T)_{\text{PST}}$ as a measure of the deviation of the PST result from the “true” rate constant which then may be used to deduce information about the anisotropy of the potential.⁴⁰ Owing to the uncertainty of the experimental temperature coefficient only the value of Φ_r (300 K) is reported, largely in agreement with other determinations.

Appendix. The symmetric top K-rotor

The rotational energy of a symmetric top is given by

$$E_r(J, K) = J(J+1)B + (A-B)K^2 \quad (30)$$

where A, B are rotational constants and J, K are quantum numbers. K is restricted to integral values between $-J$ and

+J, zero included. The J -resolved partition function of a prolate top ($A > B$) is obtained by integrating over all K :

$$Q_r(J) = (2J + 1)e^{-J(J+1)B/kT} \int_{-J}^{+J} e^{-K^2(A-B)/kT} dK \quad (31)$$

where the integral can be considered as the (semi-classical) partition function $Q_{rK}(J)$ of the prolate K -rotor for specified J . It can be evaluated analytically by means of the error function:

$$Q_{rK}(J) = \left(\frac{\pi kT}{A-B}\right)^{1/2} \operatorname{erf}\left\{\left(J + \frac{1}{2}\right)\left(\frac{A-B}{kT}\right)^{1/2}\right\} \quad (32)$$

For large x we have $\operatorname{erf}(x) \rightarrow 1$, so that $Q_{rK}(J) \rightarrow (\pi kT/A-B)^{1/2}$ for large J , which is the standard partition function for an unconstrained 1-dimensional rotor. When x is small, $\operatorname{erf}(x) \rightarrow 2x/\pi^{1/2}$, so that in order to ensure that $Q_{rK}(J) \rightarrow 1$ at $J \rightarrow 0$, $J + \frac{1}{2}$ is written for J in eqn. (32).

The similarity with eqn. (20) is obvious, with a different definition of the parameters; thus $a = (A-B)^{1/2}$, $b = [\pi/(A-B)]^{1/2}$, $r = 1$ and $y_m(J) = 0$. Hence the S-D routine described in eqns. (23)–(27) can be used as is with properly redefined parameters in $Q_{xi}(J)$.

An oblate top corresponds to $A < B$, with energy levels again given by eqn. (30), except that now $A-B$ is negative, so that each K reduces the J -rotor energy, with the result that $E_r(J, K)$ may become negative. Therefore in addition to the limitation $-J \leq K \leq +J$, the quantum number K is subject to the additional constraint by the requirement that $E_r(J, K)$ remain positive, i.e. that $K_{\max}$ be the integer nearest

$$K_{\max} = \pm \sqrt{\frac{J(J+1)B}{B-A}} \quad (33)$$

If we write $J(J+1)/(1-A/B)$ for the term under the square root, we have always $(1-A/B) < 1$ in an oblate top, so that eqn. (33) is only a very weak constraint. The K -rotor partition function for specified J is then similarly (taking $K_{\max} \cong J$)

$$\begin{aligned} Q_{rK}(J) &= \int_{-J}^{+J} e^{K^2(B-A)/kT} dK \quad (\text{oblate}) \\ &= 2\left(\frac{kT}{B-A}\right)^{1/2} \operatorname{daw}\left\{\left(J + \frac{1}{2}\right)\left(\frac{B-A}{kT}\right)^{1/2}\right\} \end{aligned} \quad (34)$$

where $\operatorname{daw}(z)$ is the modified Dawson's integral. Again this equation is similar to eqn. (20) except that $\operatorname{erf}(z)$ is replaced by $\operatorname{daw}(z)$ which can be easily obtained by a trivial modification of the $\operatorname{erf}(z)$ routine, as shown below:

The error function is defined by (ref. 9, p. 297, item 7.1.1):

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-x^2) dx = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{n!(2n+1)} \quad (35)$$

where the normalization constant is chosen so that $\operatorname{erf}(z \rightarrow \infty) = 1$. The modified Dawson's integral required here (cf. ref. 9, p. 298, items 7.1.17 and 7.1.18) is a trivial modification of the above expansion of the error function:

$$\operatorname{daw}(z) = \int_0^z \exp(x^2) dx = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{n!(2n+1)} \quad (36)$$

For small z , $\operatorname{daw}(z) \rightarrow z$, so that when $J \rightarrow 0$, the oblate top partition function $Q_{rK}(J)$ of eqn. (34) approaches unity, as required. At large z , $\operatorname{daw}(z)$ increases without limit which will cause problems if the restriction represented by eqn. (33) is not heeded. Again the S-D routine of eqns. (23)–(27) applies essentially as is by merely redefining the parameters in $Q_{xi}(J)$ and substituting the function $\operatorname{daw}(z)$ for $\operatorname{erf}(z)$.

The results obtained by inversion of $Q_{rK}(J)$ are identical to and supercede a more cumbersome procedure published previously.⁴²

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